



Science

Stage 8

Paper 2

2022

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

INSTRUCTIONS

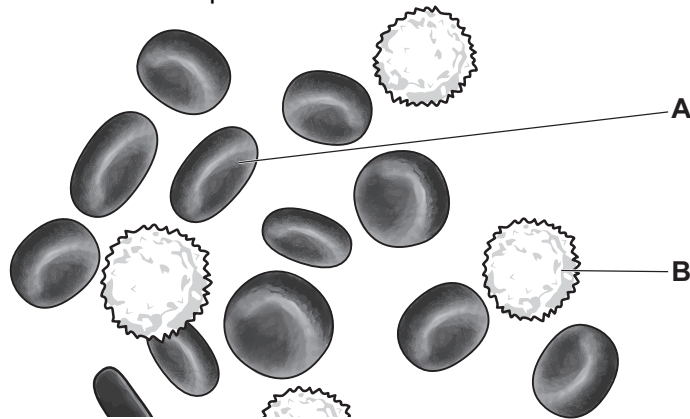
- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Safia uses a microscope to study a sample of blood from a healthy person.

Look at the view from the microscope.



- (a) Write down the function of cell A.

.....
 [1]

- (b) Safia then studies a blood sample from someone recovering from an infection.

This blood sample looks different to the sample from a healthy person.

Describe how the blood sample from a person recovering from an infection looks different.

.....

 Explain your answer.

.....
 [2]

- (c) Blood cells are transported in a liquid.

- (i) Write down the name of this liquid.

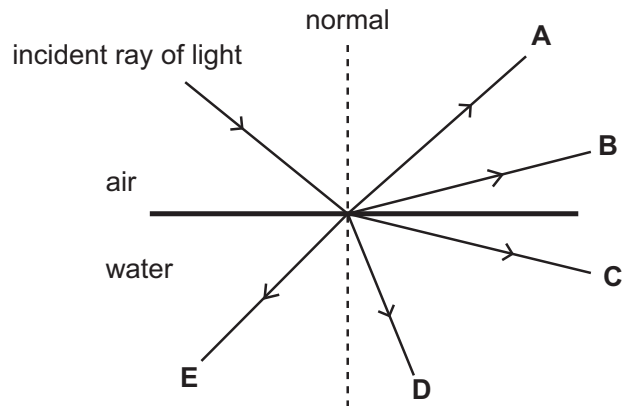
..... [1]

- (ii) Write down **one other** function of this liquid.

.....
 [1]

2 Light is reflected and refracted by water.

The diagram shows some paths a ray of light may take.



(a) Which letter shows the **reflected** ray?

Circle the correct answer.

A B C D E [1]

(b) The law of reflection is about the angle of incidence (i) and the angle of reflection (r).

Write down the law of reflection.

.....
 [1]

(c) Which letter shows the **refracted** ray of light?

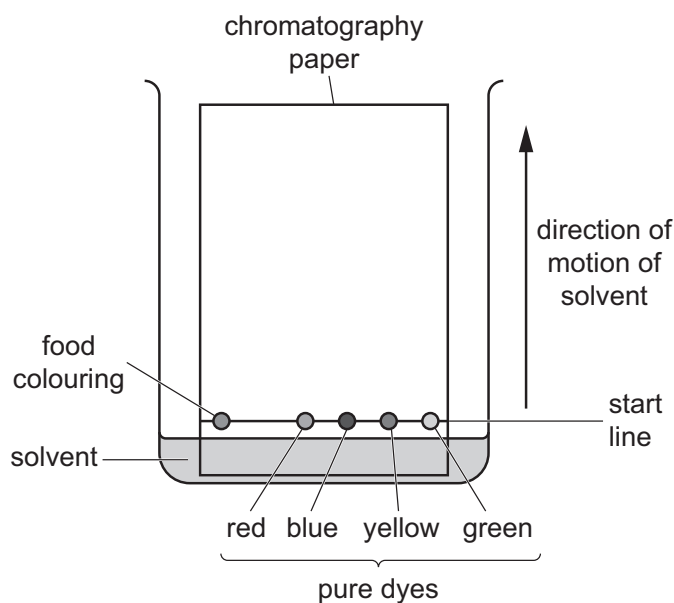
Circle the correct answer.

A B C D E [1]

3 Mike investigates the coloured dyes used to make a food colouring.

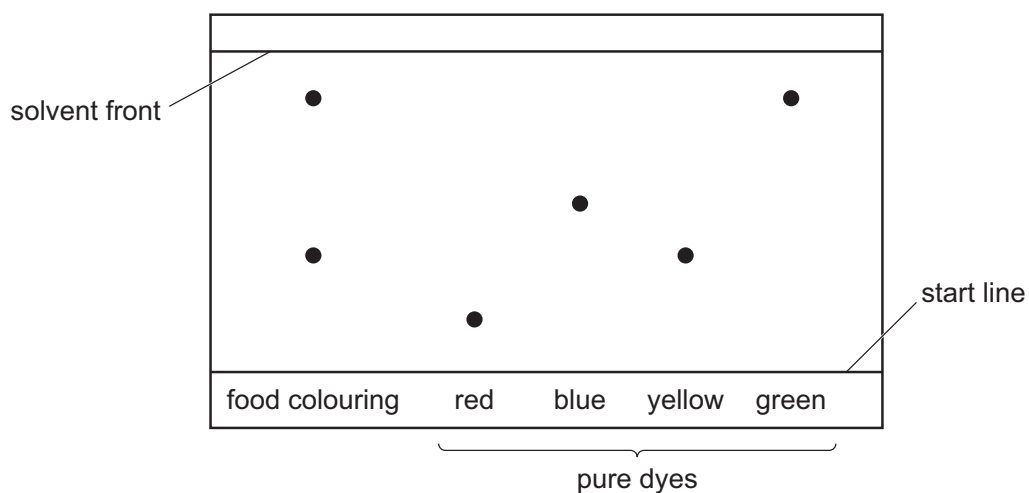
He:

- uses paper chromatography
- draws a start line in pencil on the chromatography paper
- places a dot of food colouring on the start line
- places dots of pure dyes on the start line
- puts the chromatography paper in the solvent as shown in the diagram



- waits until the solvent has soaked up the chromatography paper.

Look at Mike's chromatogram.



- (a) Mike puts his food colouring and the pure dyes on a **pencil line** rather than a line drawn in ink.

Explain why.

.....
..... [1]

- (b) Paper chromatography separates the colours in the food colouring.

Describe how.

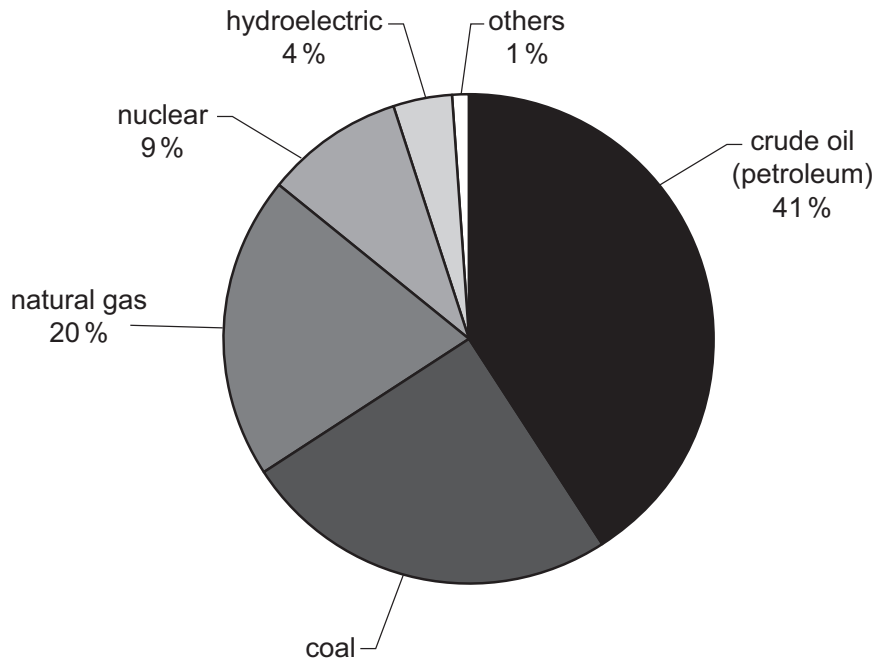
.....
..... [1]

- (c) Which coloured dyes are present in the food colouring?

.....
Explain your answer.
.....
.....
..... [2]

4 Electricity is generated using different resources.

This pie chart shows the percentage of electricity generated in a country using different resources.



(a) What percentage of the electricity is generated using coal?

..... % [1]

(b) What type of resource are crude oil (petroleum), coal and natural gas?

..... [1]

(c) Write down **two other** energy resources **not** shown in the pie chart.

1

2

[2]

- 5 Gabriella makes some sugar.

It is important that the sugar is very pure.

- (a) What is meant by a pure substance?

.....
..... [1]

- (b) Gabriella makes 12.0 g of a solid.

The solid contains 10.8 g of sugar.

What is the percentage purity of sugar in the solid?

Use the equation shown.

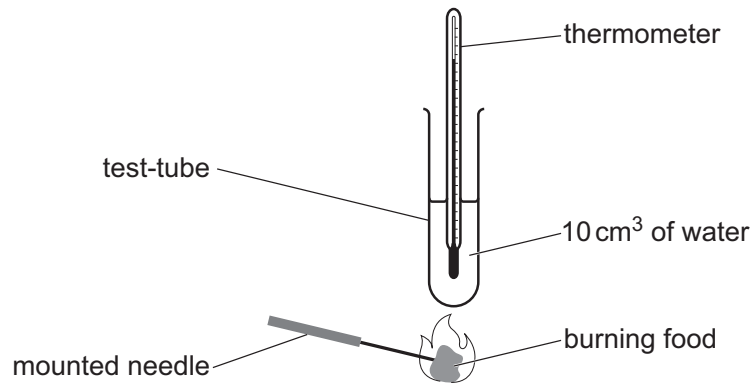
$$\text{percentage purity} = \frac{\text{mass of useful product}}{\text{total mass of solid}} \times 100$$

percentage purity = % [2]

- 6 Oliver investigates the amount of energy in different foods.

Look at the diagram.

It shows the equipment he uses.



Oliver:

- measures 1 g of food
- puts 10 cm³ of water into a test-tube
- measures the starting temperature of the water
- sets the food on fire
- holds the burning food under the test-tube until the food stops burning
- measures the final temperature of the water.

Oliver repeats the experiment using different foods.

- (a) Explain why Oliver uses the same mass of food each time.

.....
 [1]

- (b) Write down **one other** variable Oliver keeps the same.

..... [1]

Oliver writes his results in this table.

.....		
.....		
nut	28	64
biscuit	26	40
cereal	26	44
bread	27	38

(c) Complete the table by writing the headings in the table. [2]

(d) Write down the name of the food that contains the **most** energy.

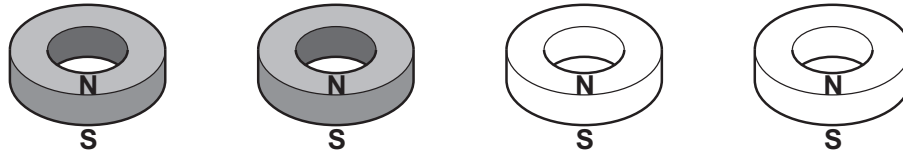
..... [1]

(e) Write down the name of the **nutrient** in food which provides a large amount of energy.

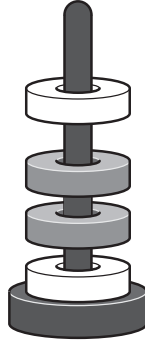
..... [1]

- 7 Ring magnets have a north pole, **N**, and a south pole, **S**.

Chen has a toy made with four ring magnets.



- (a) Chen wants to use the four ring magnets to make the pattern in the diagram.



Explain how he does this.

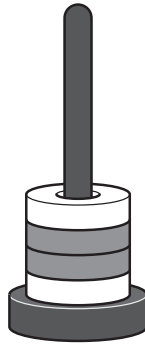
Use ideas about north poles and south poles.

.....

.....

..... [2]

(b) Chen now wants to use the four ring magnets to make a different pattern.



Explain how he does this.

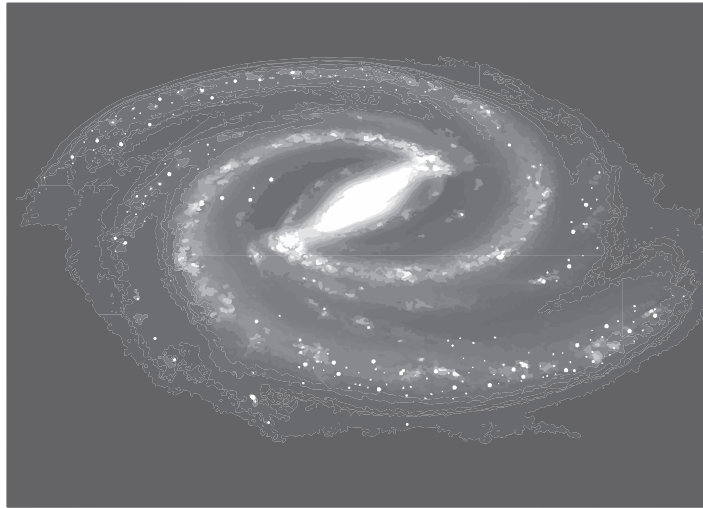
Use ideas about north poles and south poles.

.....

.....

..... [1]

- 8 Look at the picture of a galaxy.



- (a) Write down **two** types of objects that make up a galaxy.

1

2

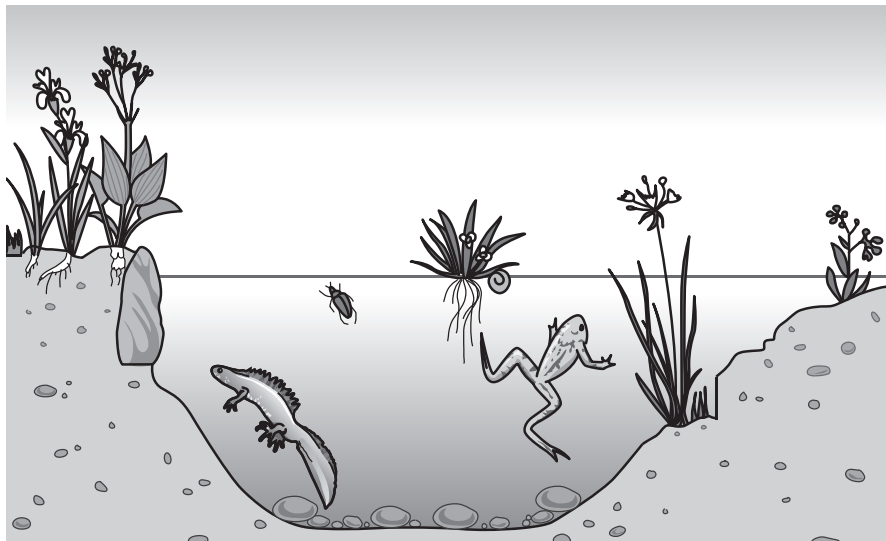
[1]

- (b) Write down the name of a piece of equipment scientists use to study galaxies.

.....

[1]

9 Look at the picture of a pond.



(a) Circle the word that describes all the living and non-living things in the pond.

abiotic

biotic

ecosystem

food web

[1]

(b) Write down **two** habitats in the picture.

1

2

[2]

10 This question is about reactions of metals.

(a) Magnesium burns in oxygen to make a new substance.

Write the **word** equation for this reaction.

..... [1]

(b) Sodium reacts vigorously with water.

Write down **two** observations you make when sodium reacts with water.

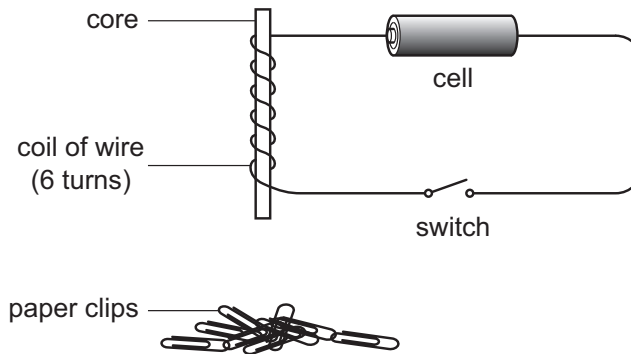
1

2

[2]

11 Aiko makes an electromagnet.

Look at the diagram of her equipment.



Aiko uses the electromagnet to pick up paper clips.

She makes this prediction,

‘Increasing the number of turns of wire around the core will make a stronger electromagnet.’

Look at Aiko’s results.

number of turns	number of paper clips picked up
2	1
4	3
6	5
8	6
10	7

(a) Is Aiko’s prediction correct?

.....

Explain your answer using her results.

.....

.....

[1]

(b) Mia tells Aiko that she needs to improve her investigation.

Describe how Aiko improves her investigation.

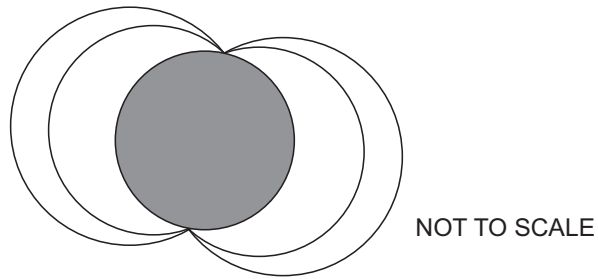
.....

Explain your answer.

.....

[2]

12 Yuri has drawn this diagram to show the magnetic field lines around the Earth.



(a) Complete the sentence below to explain why the Earth has a magnetic field.

The of the Earth acts as a [1]

(b) Name the piece of equipment that is used to show the direction of the magnetic field.

..... [1]

(c) Draw arrows on **each** of the magnetic field lines to show the direction of the magnetic field. [1]

13 Ahmed investigates the solubility of potassium nitrate in water at different temperatures.

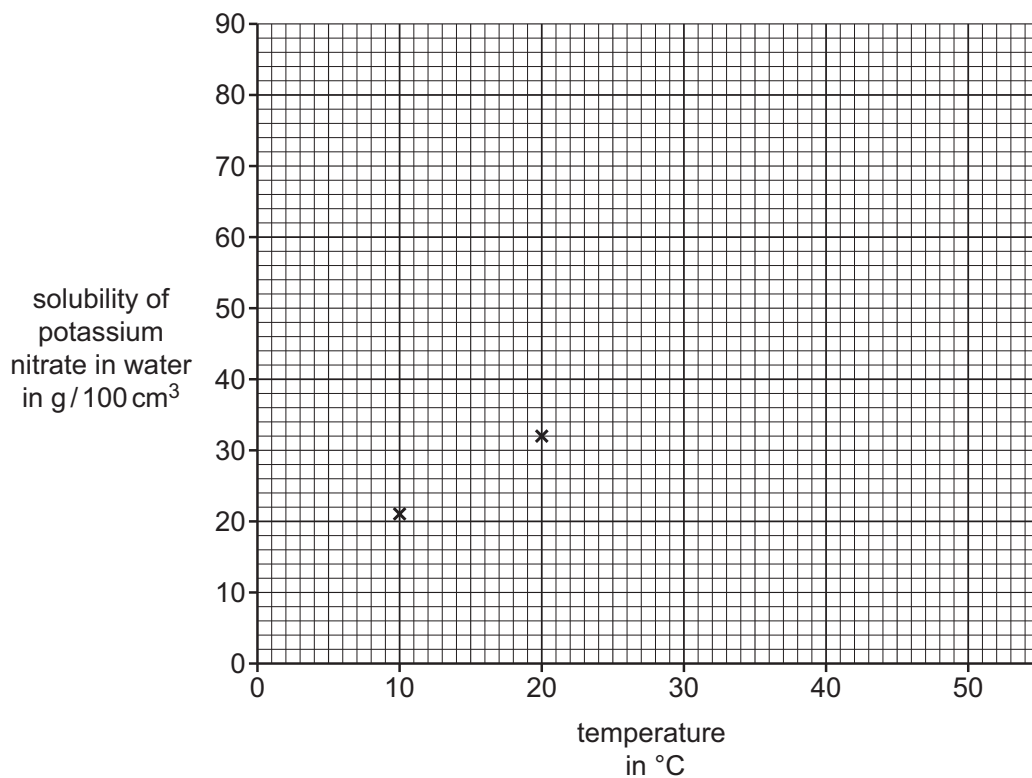
Look at his results.

temperature in $^{\circ}\text{C}$	solubility of potassium nitrate in water in $\text{g} / 100 \text{ cm}^3$
10	21
20	32
30	46
40	64
50	86

(a) Plot Ahmed's results on the grid.

Two points have been done for you.

Draw the curve of best fit through the points.

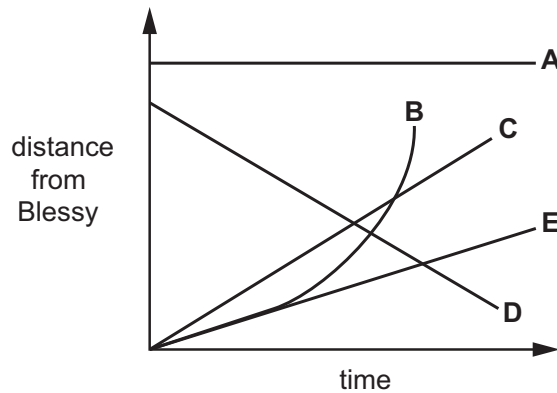


[2]

(b) Complete the sentence.

As the temperature increases, the solubility of potassium nitrate in water [1]

14 Blessy draws five different distance / time graphs on the same axes.



The distance axis shows how far an object is away from Blessy.

(a) Which graph shows a **stationary** object?

Circle the correct answer.

A B C D E [1]

(b) Which graph shows an object moving **towards** Blessy?

Circle the correct answer.

A B C D E [1]

(c) Which graph shows an object getting **faster**?

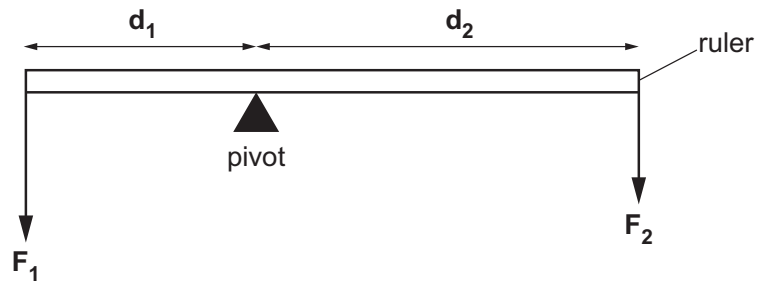
Circle the correct answer.

A B C D E [1]

15 Look at the diagram.

It shows two different forces F_1 and F_2 acting on a ruler.

The ruler is balanced.



Explain why the ruler is balanced.

Use ideas about the principle of moments.

.....

.....

..... [2]

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The Periodic Table of Elements

Group																	
1	2	1 H hydrogen 1										3	4	5	6	7	8
		Key atomic number atomic symbol name relative atomic mass															
3 Li lithium 7	4 Be beryllium 9											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —
lanthanoids		57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175	
actinoids		89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —	